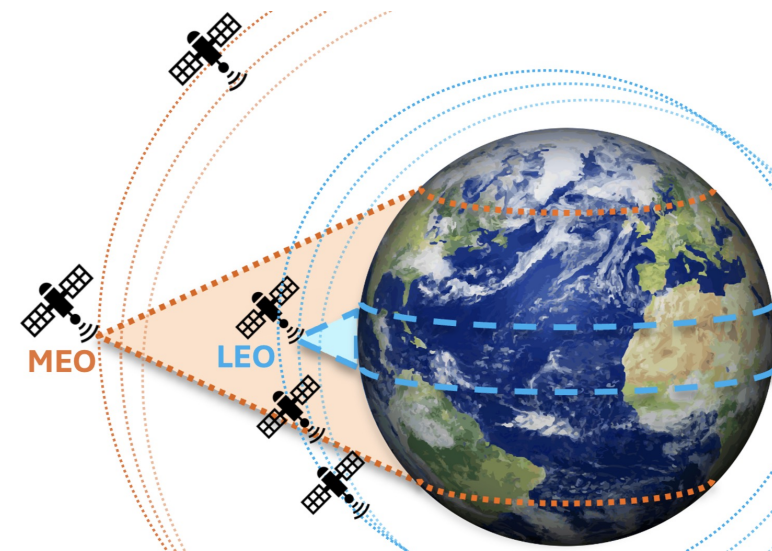


# Starlink Beacons for Passive LEO-Aided 9D Navigation



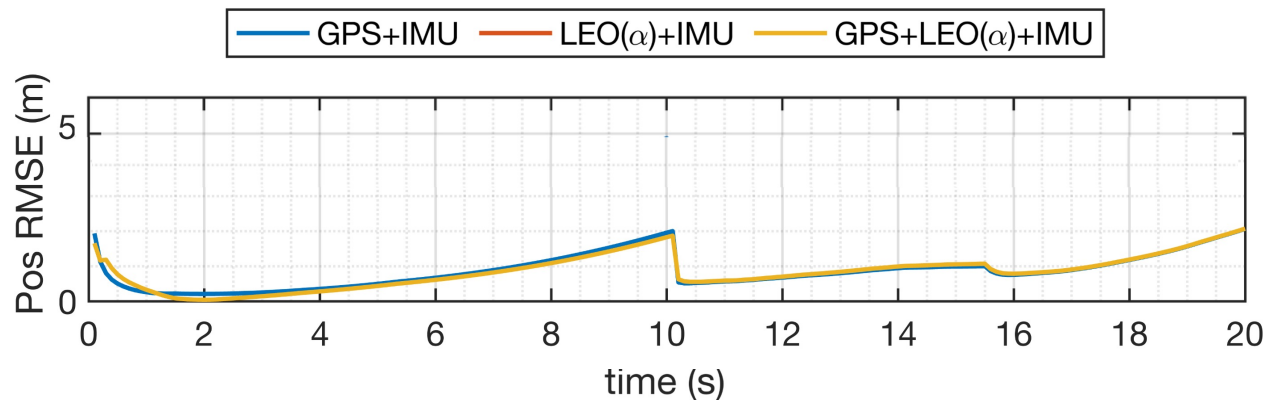
Nisanur Camuzcu, Tiep M. Hoang, Alireza Vahid

09/08/2026

# Motivation: fragile dependence on GNSS

Why another navigation source if GPS already works?

- An **independent** and **passive** source when GPS is unreliable



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Why another navigation source if GPS already works?

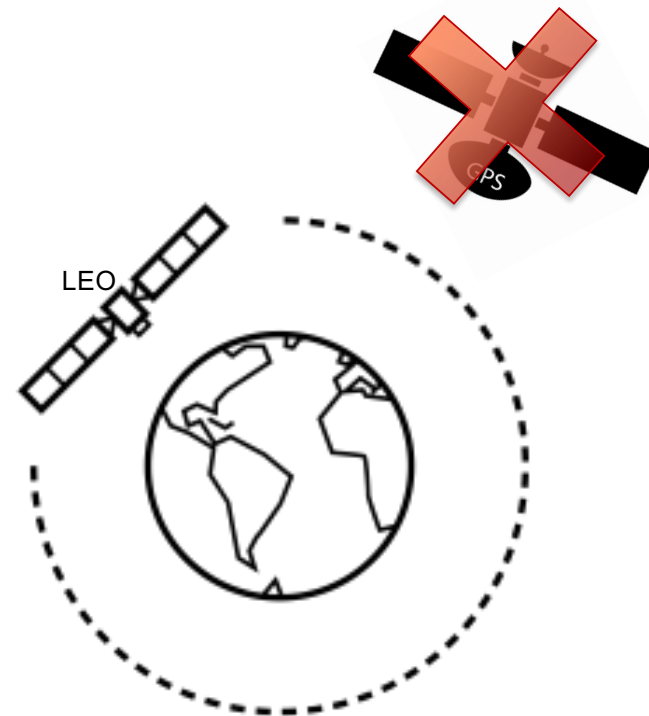
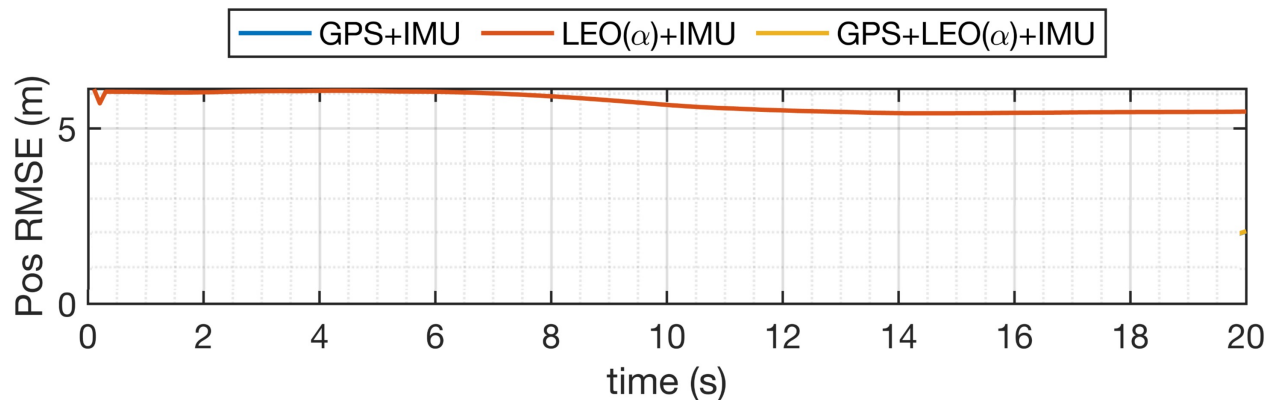
- An **independent** and **passive** source when GPS is unreliable



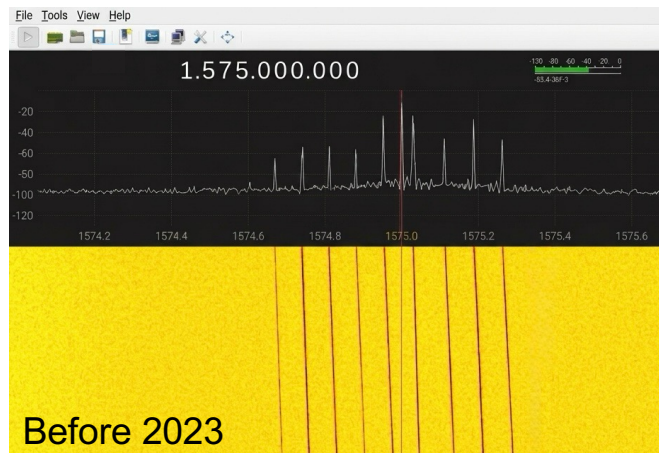
# Motivation: fragile dependence on GNSS

Why another navigation source if GPS already works?

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# Background and Gap

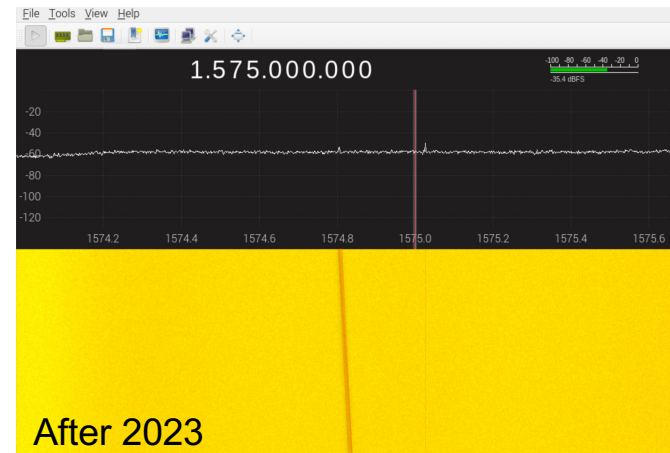


<https://forum.amsat-dl.org/index.php?thread/3981-receiving-starlink-satellite-beacons/>



single LNB  
with 60 dB gain

**Pilot tone beacons**  
reduced power in 2023 [1]



**Can passive beacons still be used and support full 9D navigation, when integrated with IMU and GPS?**

3D position + 3D velocity + 3D attitude

**Tradeoff:** richer measurements [2]–[5] vs. simpler access [6], [7]

# Main Contribution

- We develop and evaluate an **end-to-end passive LEO-aided 9D navigation pipeline using Starlink pilot-tone beacon Doppler-rate**.

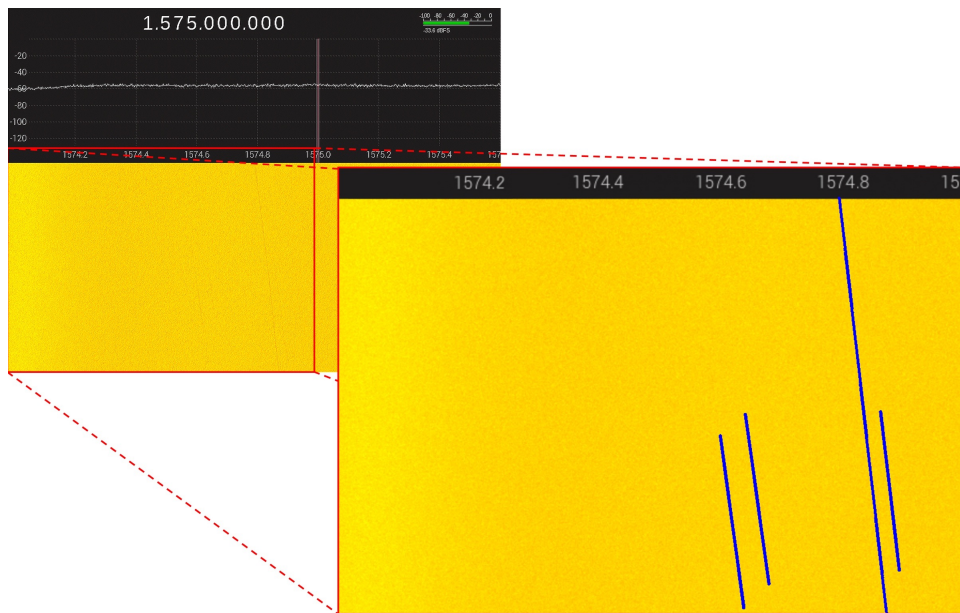


- Key questions:

**Can passive beacon Doppler-rate support 9D navigation, and how does it scale?**

**How does its contribution compare with a range-type LEO observable?**

# Passive Observable: Doppler-rate from pilot tone beacons

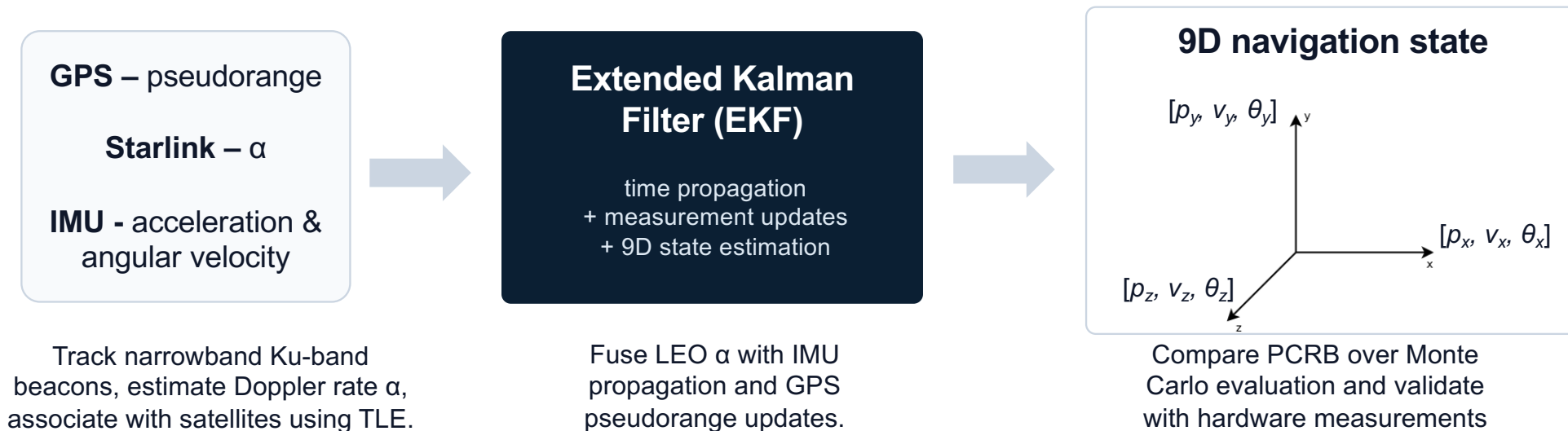


$$\alpha_s(t) = \dot{f}_D(t) \approx \frac{f_c}{c} \ddot{\rho}$$



- Fast LEO motion creates satellite-specific Doppler-rate signatures.
- The observable is lightweight: no payload decoding is required [6].
- Accuracy depends on tracking, timing, oscillator stability, and association quality.

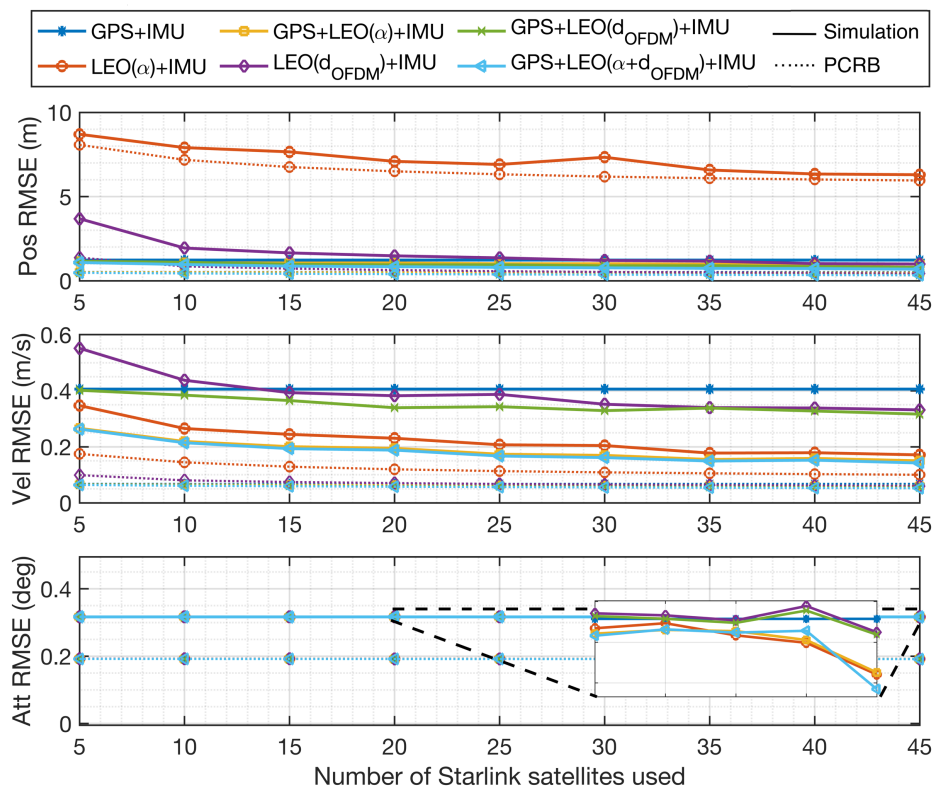
# Fusion and 9D Navigation



**What navigation value do we gain from an easy-to-observe passive  $\alpha$  measurement, and what do we lose relative to range?**



# Simulation: What Does Passive $\alpha$ Contribute?



LEO( $\alpha$ )+IMU: roughly 6-9 m position RMSE

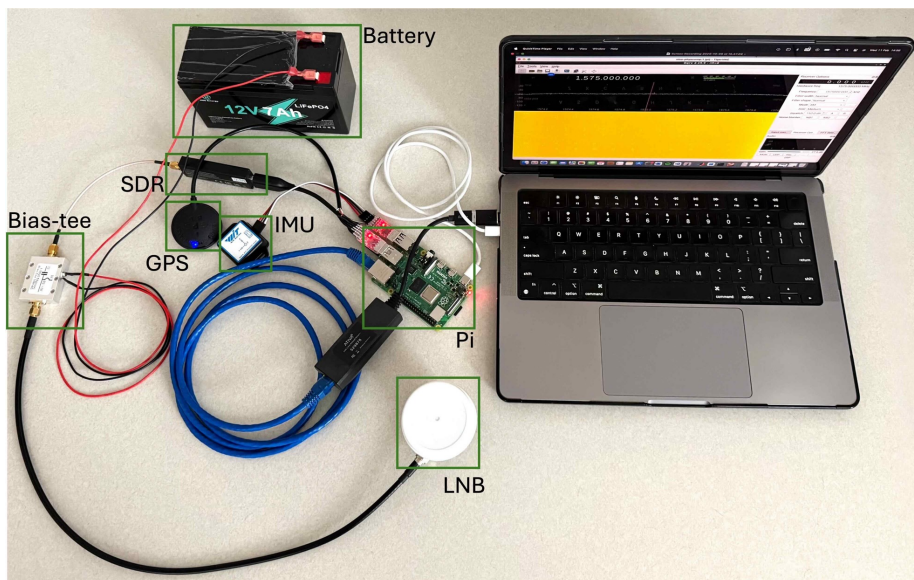
GPS+LEO( $\alpha$ )+IMU: trends toward GPS-driven 1-2 m regime

Range-type LEO baseline gives stronger translation gains; combining observables is best.

**Doppler-rate  $\alpha$  is a robust, passive aiding measurement; not a substitute for range information.**

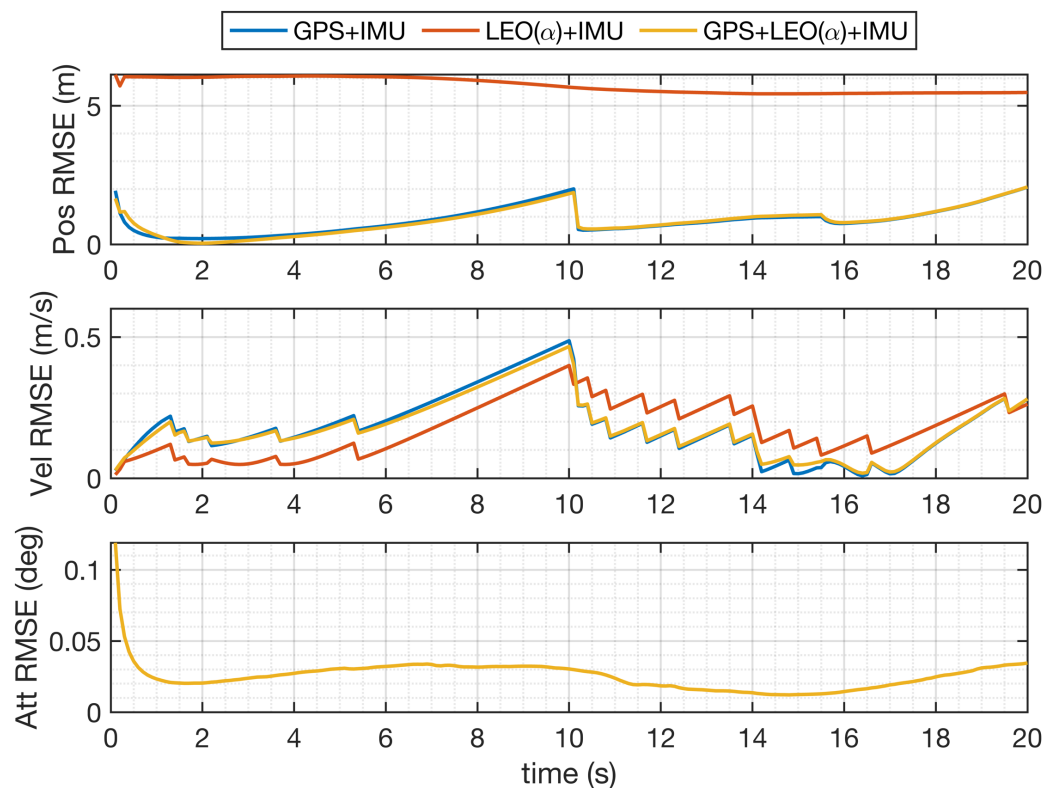
# Hardware Validation with Real Starlink Beacons

- Ku-band pilot-tone beacon capture using LNB + RTL-SDR V3



- u-blox GNSS + WitMotion IMU sensors at 10 Hz
- 20 s synchronized capture
- 5 Starlink  $\alpha$  measurements associated and fused

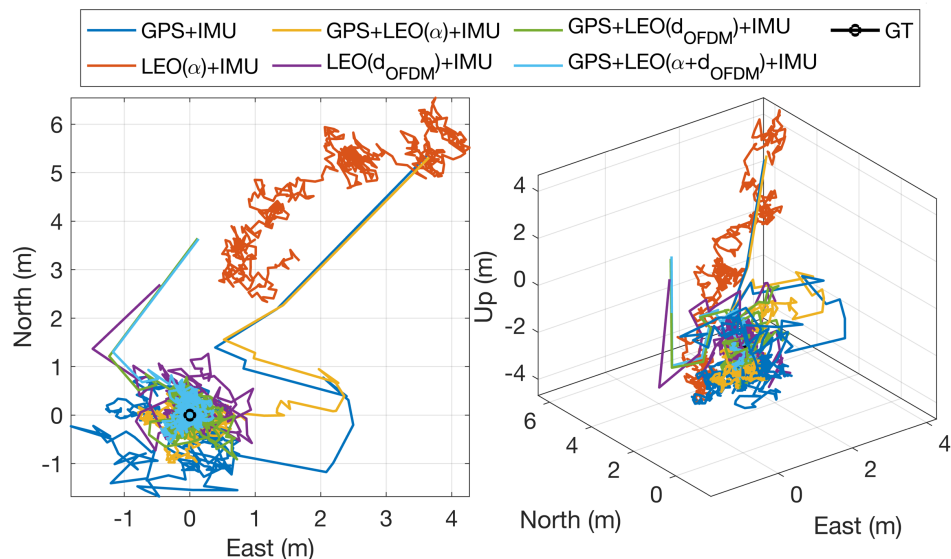
# Hardware: passive $\alpha$ gives bounded errors + stable velocity



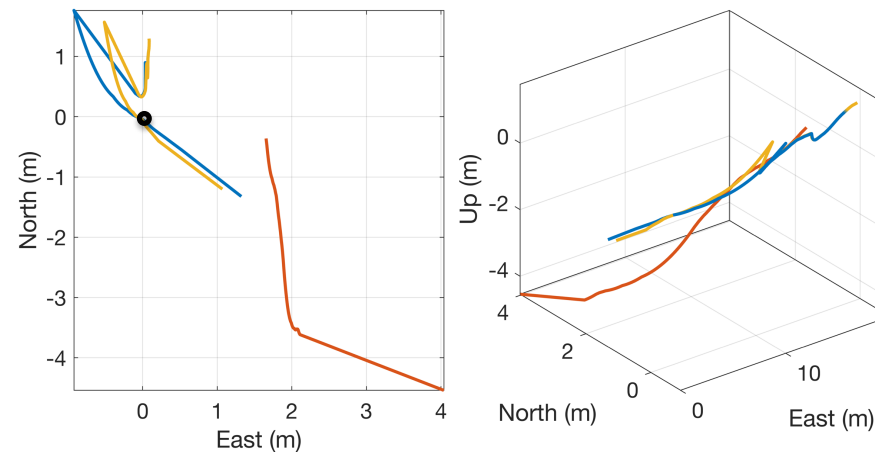
- LEO( $\alpha$ )+IMU remains bounded at several-meter position error.
- GPS+IMU reaches ~1 m accuracy under nominal GPS.
- Hybrid solution closely follows GPS when GPS is strong and adds an independent transition aid.

**Pilot-tone beacon  $\alpha$  is useful LEO aiding, even with limited updates.**

# Trajectory view: $\alpha$ -only drifts, hybrid fusion stays anchored



Simulation trajectories



Hardware trajectories

- $\alpha$ -only can exhibit bias or drift but maintains a usable navigation solution.
- GPS-aided solutions remain concentrated near the reference.

**LEO  $\alpha$  constrains the navigation solution; hybrid fusion keeps the trajectory anchored.**

# Takeaways

## Implication

### **Passive, complementary, scalable**

LEO beacons can provide an independent aiding channel for navigation, and more visible LEO satellites improve measurement diversity.

## Tradeoff

Doppler-rate is directly observable with lower receiver complexity, while range provides stronger absolute positioning information.

## Validation scope

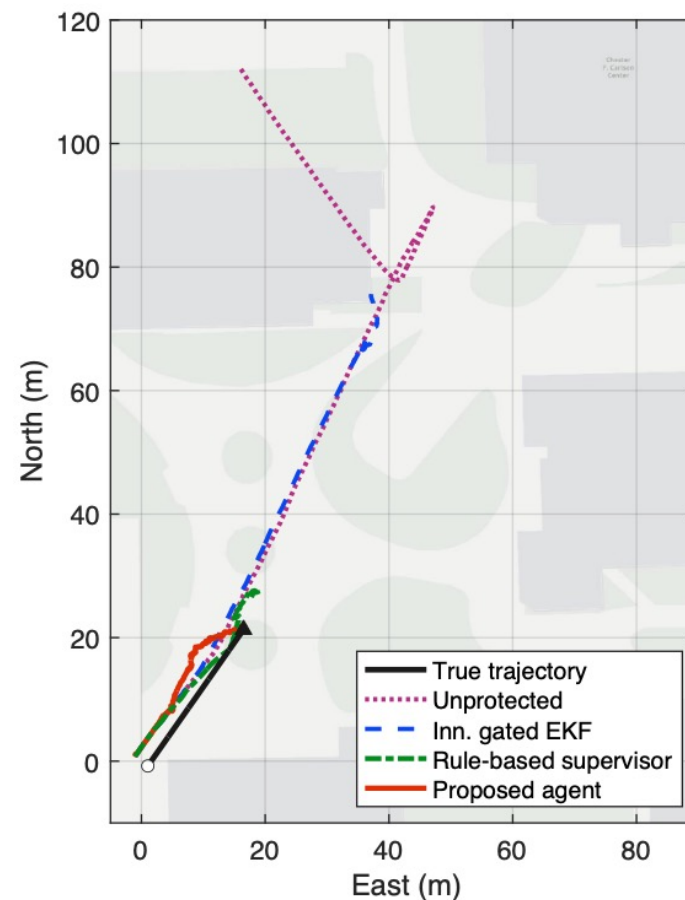
Initial hardware validation uses a 20 s capture with five Starlink  $\alpha$  updates; longer field tests and multi-constellation validation are needed.

**The concept extends to other LEO constellations whenever a deterministic downlink feature like pilot-tone beacon is observable.**

# Conclusion

Future work will focus on improving performance with

- longer field tests,
- multi-constellation generalization, and
- agentic AI for robustness and adaptive trust.



# References

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# Thank you!

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